

# CHEM 396 Interim Report

Simulation of Amorphous Silicon using Machine-Learning Interatomic Potentials

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## Abstract

This report covers the first six weeks (June 15 to July 27, 2026) of a project to build models of amorphous silicon by melt-quench molecular dynamics using Moment Tensor Potentials. The main deliverable so far is a working toolchain on Narval: two custom LAMMPS builds with the ML-MTP package, one on CPU and one on GPU through Kokkos, both of which required debugging of the supplied build scripts before they ran. With that in place, a 216-atom system was used to characterize the supplied potentials, the undocumented atom-type mapping was identified, the equilibrium lattice constant was measured at 0.8% above the experimental value, and the melt-stable and pruned potentials were found to prefer different volumes, a mismatch that running under NPT throughout should absorb. Parallelization and a cheaper pruned potential together cut the projected cost of the full protocol to a single overnight run. A 30 ps ramp to 2500 K gave a confirmed liquid, and the 1 ns hold is currently executing. The quench to 300 K, the structural analysis, the comparison against Zongo et al. (2025) and experiment, and the application of MAP remain to be done.

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# 1 Introduction

Amorphous silicon (a-Si) lacks the long-range order of crystalline silicon (c-Si). Most atoms are still bonded to four neighbours in roughly tetrahedral arrangements, but bond lengths and angles vary rather than being fixed. a-Si is used in solar cells, thin-film transistors, and battery anodes, and its properties depend strongly on coordination defects, where atoms have too few or too many bonds.

Simulating a-Si requires a model of the forces between atoms. Density functional theory (DFT) is accurate but expensive, and empirical potentials are fast but limited. Machine-learning potentials sit in between: they learn the energy from DFT data. This project uses one such method, the Moment Tensor Potential (MTP).

## 1.1 Objectives

The aim is to generate a-Si models through melt-quench molecular dynamics driven by an MTP, and to analyze the structures using the Morphological Autoregressive Protocol (MAP). Concretely:

1. Build a working MTP-enabled LAMMPS toolchain on Narval (CPU and GPU) and validate it.
2. Characterize the five supplied silicon MTPs (cost, stability, preferred volume) and settle a practical melt-quench protocol.
3. Run the melt-quench protocol and generate a-Si configurations.
4. Characterize the resulting networks (coordination, bond angles, rings, structure factor), compare against Zongo et al. (2025) and experiment, then apply MAP.

## 1.2 Status at the interim point

Objectives 1 and 2 are complete, while objective 3 is in progress: the melt has run successfully and the 1 ns hold at 2500 K is currently executing.

# 2 Background

## 2.1 Amorphous silicon

c-Si has the diamond cubic structure, with every atom bonded to four neighbours at fixed bond lengths (2.35 Å) and angles (109.5°). In a-Si the same local bonding survives, but the periodicity is gone: bond lengths and angles spread around their crystalline values, and a small fraction of atoms are coordination defects, either dangling bonds (threefold) or floating bonds (fivefold). This makes a-Si slightly softer (bulk modulus 70 to 80 GPa vs. 98 GPa) and slightly less dense (2.28 vs. 2.33 g/cm<sup>3</sup>).

The standard structural picture is the continuous random network (CRN): a non-periodic network in which nearly every atom is fourfold coordinated, with disorder entering through geometry rather than through broken bonds. Bond angles spread around 109.5° with an RMS width near 10°. Structure beyond nearest neighbours shows up in ring statistics: in c-Si every closed loop of bonds is six-membered, while in a-Si the distribution peaks at five-, six-, and seven-membered rings, with three-membered rings absent and four-membered rings rare because both are heavily strained. There is no single agreed rule for counting rings in a disordered network (King, Guttman, and primitive definitions all exist), so any reported distribution must be read alongside the counting method used.

Defect concentration depends on how the sample was made. Experimentally, as-implanted a-Si is defective but anneals toward a lower-energy state. In simulation, melt-quench MD traps defects, more so at faster cooling, while bond-switching methods like Wooten-Winer-Weaire relax toward near-perfect fourfold coordination.

A long-standing debate asks whether real a-Si is a pure CRN or contains small crystal-like regions (the paracrystalline model). Fluctuation electron microscopy results (Treacy and Borisenko, 2012) were read as evidence for paracrystallites, and recent large machine-learning MD models (Rosset, Drabold and Deringer, 2025) also show paracrystalline order. This matters here because melt-quench MD is exactly the route reported to trap crystalline grains, so the structures generated in this project must be checked for residual crystallinity rather than assumed to be pure CRN.

## 2.2 Interatomic potentials

MD simulations advance atoms in time by integrating Newton’s equations, which requires the force on each atom at every step, and hence a model of the potential energy as a function of atomic positions. Potentials are usually classified by where their parameters come from. Ab initio methods like DFT compute the energy from quantum mechanics directly: accurate but expensive, with cost growing roughly as the cube of the number of electrons. Empirical potentials (Lennard-Jones, Stillinger-Weber, Tersoff, EAM) fit a fixed functional form to data which is cheap but limited by the rigidity of the form. A simple pair potential like Lennard-Jones cannot describe silicon at all, since silicon demands specific bond angles and LJ has no angular term. Machine-learned potentials have no fixed form; the structure-energy relationship is learned from a large set of DFT reference calculations.

## 2.3 Moment Tensor Potentials

MTPs (Shapeev, 2016) are machine-learned potentials whose accuracy can be dialed up in a controlled way by adding more basis functions, at the cost of speed (the level of the potential). They are physically consistent (invariant under rotations, reflections, and relabelling), scale linearly with the number of atoms, and handle multiple elements.

Until recently the MTP had no GPU version. Meng et al. (2026) added one inside the LAMMPS Kokkos package, a portability layer that lets one version of the code run on CPUs and different GPUs. Benchmarks on Narval show the optimized CPU variant reaching up to  $2\times$  over the older MLIP-3 code, with the GPU variants much faster and scaling to million-atom systems. The GPU variants are intended for roughly 2000 atoms or more per GPU. This Kokkos version is the one used in this project.

## 2.4 The benchmark study: ARTn-MTP models of a-Si

Zongo et al. (2025) is the central reference for this project. It combines the activation-relaxation technique (ARTn), a saddle-point search method that constructs barrier crossings deliberately instead of waiting for thermal fluctuations, with the unified Si/O/SiO<sub>2</sub> MTP of Zongo et al. (2024) to build seven a-Si models of 216 to 4096 atoms, compared against melt-quench MD with the same potential, against experiment, and against reference models from the literature. Notably, the MTP training set contained no a-Si configurations, so the study also tests transferability.

The ARTn-MTP structures reach defect levels below 2% in most models, with bond angles centred on  $109.5^\circ$  (spread near  $10^\circ$ ), ring statistics peaking at 5-, 6-, and 7-membered rings, a bulk modulus near 75 GPa, and excess energies inside the experimental window. Crystallinity, measured with OVITO’s identify-diamond-structure modifier, was zero in four models, all generated from a

random start with no MD prerun. The authors’ main conclusion is that ARTn-MTP gives both low crystallinity and low defect counts, while melt-quench MD tends to trap crystalline grains. Since this project uses the melt-quench route, the paper is both the target for structural comparison and a warning about the failure mode to check for.

## 2.5 MAP

The Morphological Autoregressive Protocol (Madanchi et al., 2024) is a generative machine-learning method that builds large models of disordered matter at a cost linear in system size. It places the structure on a grid and generates it one point at a time, with each point conditioned only on already-placed neighbours within a finite correlation length, using a PixelCNN. Tested on liquid water, it reproduces the radial distribution functions and bond-angle and bond-length distributions, and its outputs can seed stable MD runs. A follow-up (Madanchi and Simine, 2026) uses the same trained model as an anomaly detector in a supercooled Lennard-Jones glass-former: a per-particle likelihood score flags unusual local arrangements, and its dips line up with about 88% of independently identified rearrangement events. MAP will be applied to the structures generated in this project in the final stage.

## 2.6 Thermodynamic ensembles

Three ensembles matter here. NVE (constant number, volume, energy) integrates Newton’s equations with no thermostat and is a check on numerical stability, since energy should be conserved. NVT (constant number, volume, temperature) adds a thermostat and is standard for runs at fixed temperature. NPT (constant number, pressure, temperature) adds a barostat that lets the box change size, and is the only ensemble in which the system can find the density the potential actually prefers. Running NVT at the wrong volume forces a compressed or stretched state, visible as a persistent non-zero pressure. This becomes the central issue in Section 4.6.

## 2.7 Software

LAMMPS is the standard open-source MD engine used here. It reads a plain-text input script listing the system, the potential, and the simulation steps. The potential is a swappable module selected with `pair_style`; since the mtp styles are not in the standard distribution, LAMMPS must be compiled from source with the ML-MTP package (Section 3.2). Work runs on Narval, a Calcul Québec/DRAC cluster, accessed with PuTTY (SSH) and WinSCP (file transfer) from Windows. Jobs go through the SLURM scheduler, either as batch scripts (`sbatch`) for production runs or as interactive allocations (`salloc`) for tasks needing trial and error, such as compiling. Structures and trajectories are visualized in VMD; data files load through the TopoTools plugin in the Tk Console, trajectories through the normal loader.

# 3 Methodology

## 3.1 Toolchain validation: Lennard-Jones melt

Before touching silicon, a standard Lennard-Jones melt was run to confirm the full pipeline (job submission, LAMMPS execution, output, visualization) worked. The system is 13,500 atoms on an fcc lattice at reduced density 0.8442, `pair_style lj/cut 2.5`, initial reduced temperature 1.44, integrated in NVE for 10,000 steps. This test case was chosen because it is the standard

LAMMPS example, so the expected behaviour is well known. The same input was later reused as the acceptance test for both custom builds.

### 3.2 Building LAMMPS with the MTP

The MTP-enabled code was provided by Alliance support as an archive (`LMP_MTP.tar.gz`) bundling the LAMMPS source, the MTP sources, and two build scripts (CPU and GPU/Kokkos). Compilation was done on a compute node through an interactive allocation, after loading the required modules (`gcc`, `openmpi`, `cmake`, `cuda`, and supporting libraries). Both scripts needed corrections before producing working binaries; since diagnosing them was a substantive part of the work, they are reported as results in Section 4.2.

### 3.3 Simulation protocol

Five silicon MTPs were provided: two melt-stable potentials (levels 22 and 26), fitted to stay robust at melt temperatures, and three pruned potentials (`pruned_1` to `pruned_3`), cheaper but potentially unstable in a melt. The melt-quench protocol is:

1. **Melt.** Heat a c-Si lattice to about 2500 K with a melt-stable potential, and save a snapshot.
2. **Hold.** Hold the melt at temperature for 1 ns to erase memory of the crystal.
3. **Hand off.** Reload the snapshot with a pruned potential; the atoms are unchanged, only the force model changes.
4. **Quench.** Cool from the melt temperature to 300 K, fast enough that the liquid structure freezes in as a-Si rather than recrystallizing.

The system throughout is a 216-atom diamond lattice ( $3 \times 3 \times 3$  cells) built at the crystalline lattice constant 5.431 Å, periodic boundaries, `units metal`, a 1 fs timestep, and initial velocities at 300 K. This matches the smallest models of Zongo et al. (2025), allowing direct comparison.

LAMMPS has no explicit cooling-rate command; `fix npt` ramps linearly between a start and target temperature over the run, so the rate is set by the run length:

$$\text{rate} = \frac{\Delta T}{N_{\text{steps}} \times \Delta t}. \quad (1)$$

At the benchmark rate of 1 K/ps, cooling from 2500 K to 300 K takes  $2.2 \times 10^6$  steps.

Zongo et al. only ever mention NPT, so NPT is used throughout. As discussed in Section 5.3, this also absorbs the volume mismatch between the two potential families automatically.

## 4 Results

### 4.1 Lennard-Jones validation

On the stock module build, the total energy per atom stayed essentially constant at  $-4.62$  (reduced units) through the NVE run, confirming stable integration, while the temperature relaxed from 1.44 to about 0.70 as the lattice melted. The run took about 95 seconds of wall time with no dangerous neighbour-list builds. The trajectory was transferred with WinSCP and loaded in VMD, which shows the fcc lattice melting into a disordered fluid (Fig. 1).

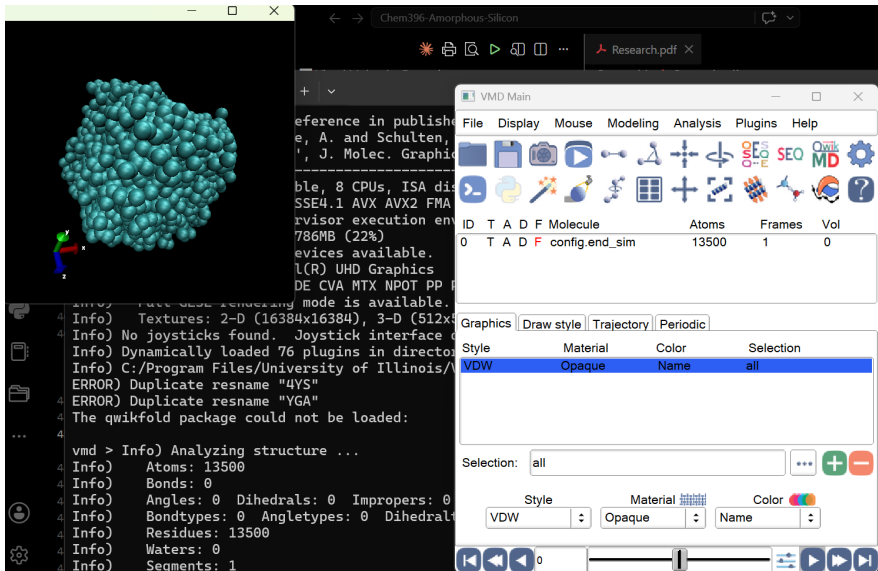


Figure 1: The Lennard-Jones melt in VMD, equilibrated into a disordered fluid.

## 4.2 Diagnosis and correction of the MTP builds

Neither supplied build script produced a working binary as distributed. Five faults were identified and fixed (Table 1).

For the CPU build: the binary crashed with “Illegal instruction” because it was compiled for the wrong CPU architecture (fixed with `-march=core-avx2`); an inverted conditional in the copy block silently skipped the MTP source files, producing a binary without MTP support; and the archive contained stale nested CMake caches from a previous build, which `--fresh` does not reset, so the whole `_build` directory had to be deleted before rebuilding. After a clean rebuild, `lmp -h` listed ML-MTP with the `mtp` pair styles available.

For the GPU build: the KOKKOS source folder was never copied, so the `mtp/kk` styles were absent; the CUDA backend of Kokkos was never enabled (fixed with the `kokkos-cuda` preset and `-DKokkos_ARCH_AMPERE80=ON` for Narval’s A100s); and two options had to be removed, since `-march=core-avx2` breaks the NVIDIA compiler and the OpenCL GPU package fails and is redundant next to Kokkos.

| Symptom                                | Cause                                    | Fix                                                                 |
|----------------------------------------|------------------------------------------|---------------------------------------------------------------------|
| Illegal instruction (core dumped)      | Wrong CPU architecture                   | Add <code>-march=core-avx2</code> (CPU build only)                  |
| ML-MTP absent from <code>lmp -h</code> | Inverted conditional skipped source copy | Correct the <code>if [ ! -d ... ]</code> logic                      |
| Rebuild produced no new binary         | Stale nested CMake caches                | Delete <code>_build</code> entirely                                 |
| <code>mtp/kk</code> styles absent      | KOKKOS folder never copied               | Copy it into <code>src/KOKKOS/</code>                               |
| Binary would not run on GPU            | CUDA backend not enabled                 | <code>kokkos-cuda</code> preset + <code>Kokkos_ARCH_AMPERE80</code> |

Table 1: Build faults in the supplied LAMMPS-MTP archive and their fixes.

Both builds were validated on the identical Lennard-Jones benchmark (Table 2) and are fully functional on Narval.

| Build                       | Hardware      | Wall time  | Relative speed |
|-----------------------------|---------------|------------|----------------|
| <code>lammps-mtp-cpu</code> | 1 CPU core    | 4 min 16 s | 1.0×           |
| <code>lammps-mtp-gpu</code> | 1 NVIDIA A100 | 15.6 s     | ~ 16×          |

Table 2: Validation of both builds on the 10,000-step, 13,500-atom LJ melt.

### 4.3 Atom type mapping

The potential file declares two species but does not say which is which. This was settled by direct test: running the 216-atom diamond lattice as type 1 gave a stable run at  $2.27 \text{ g/cm}^3$ , close to the  $2.33 \text{ g/cm}^3$  of c-Si, while running it as type 2 gave a starting pressure of  $-2.24 \times 10^6 \text{ bar}$  and the box collapsed within a few steps. Type 1 is therefore silicon and type 2 oxygen, consistent with a potential trained on the Si/O/SiO<sub>2</sub> system of Zongo et al. (2024).

### 4.4 Equilibrium volume under the melt-stable potential

An NPT run at 300 K and 0 bar let the box relax: the volume grew from  $4325 \text{ \AA}^3$  to about  $4433 \text{ \AA}^3$ , a lattice constant of  $5.476 \text{ \AA}$ , 0.8% above the measured  $5.431 \text{ \AA}$ . This overestimate is expected from the GGA functional used to produce the training data. Pressure swung over thousands of bar, normal for 216 atoms since fluctuations scale as  $N^{-1/2}$ . A follow-up NVT run with the box fixed at the settled constant showed pressure hovering near zero and steady potential energy, confirming the system is no longer strained.

Both runs used one CPU core and took about 20 minutes for 5000 steps, or roughly 0.3 ns of simulated time per day, with over 99% of the time in force evaluation. That is fine for a 30 ps melt but hopeless for cooling: 1 K/ps from 2500 K to 300 K is 2.2 ns, about seven days on one core. This motivated the optimization work below.

### 4.5 Parallel scaling and pruned-potential cost

The same 5000-step NVT run was repeated on 1, 4, and 8 cores (Table 3 and Fig. 2). Scaling is close to linear out to 8 cores. Going beyond 8 cores was not attempted, because two costs grow as the core count rises: the fraction of time spent on communication between cores went from 0.4% at 4 cores to 4% at 8, and with only 27 atoms left per core, there is not much work remaining to split up. Adding more cores would mostly add communication overhead for little gain, so 8 cores was used for all later runs.

| Cores | Wall time | Speedup | Atoms per core |
|-------|-----------|---------|----------------|
| 1     | 22:55     | 1.0×    | 216            |
| 4     | 5:07      | 4.5×    | 54             |
| 8     | 2:43      | 8.4×    | 27             |

Table 3: Wall time for a 5000-step NVT run of 216 Si atoms with `22.almtp`.

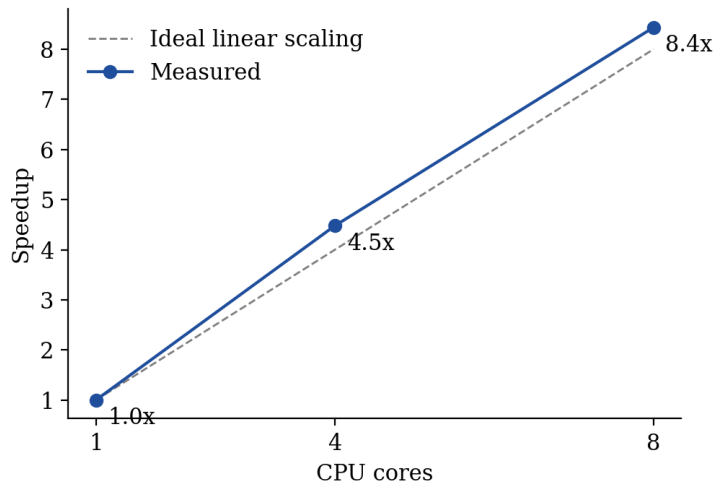


Figure 2: Parallel speedup of the 5000-step NVT run against the ideal linear case. Scaling is near-linear to 8 cores; the superlinear value at 4 cores likely reflects cache effects at small per-core atom counts.

Swapping `22.almtp` for `pruned_1.almtp` on 8 cores cut the run from 2 min 43 s to 14 s, about  $12\times$  faster (357 vs. 31 steps/s). Combined with the parallel speedup, this is roughly a hundredfold reduction over the starting point. Figure 3 summarizes both wall-time comparisons: the CPU vs GPU builds on the Lennard-Jones benchmark, and the two potentials on the silicon run.

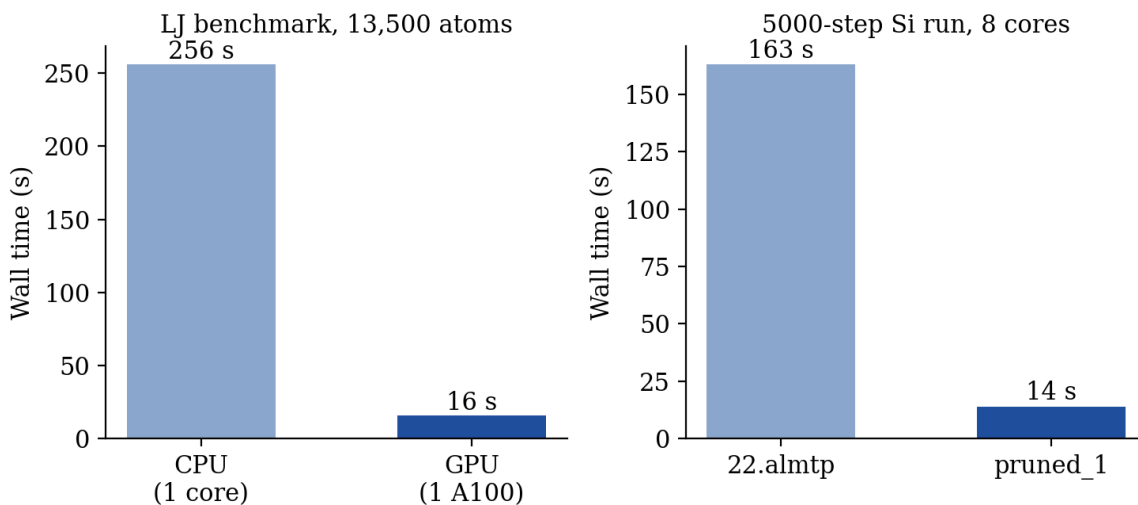


Figure 3: Wall-time comparisons. Left: CPU vs GPU builds on the Lennard-Jones benchmark (Table 2). Right: melt-stable vs pruned potential on the 5000-step silicon run.

#### 4.6 The two potentials prefer different volumes

The same test produced an unexpected result. At the box volume where `22.almtp` sits at zero pressure ( $4433 \text{ \AA}^3$ ), `pruned_1.almtp` holds a steady pressure of about  $-12,400$  bar: it wants a smaller box. This matters directly for the protocol, which switches potentials midway; the

equilibrium volume of one potential is not the equilibrium volume of the other, so the box size cannot simply be carried across the handoff. The implications are discussed in Section 5.3.

#### 4.7 Projected cost of the full protocol

Combining the throughput numbers with the step counts from Eq. 1 gives Table 4. The 1 ns hold dominates, despite fewer steps than the cooling, because the melt-stable potential is about twelve times more expensive per step. These figures assume the per-step cost measured on a static 300 K crystal; a hot liquid triggers frequent neighbour-list rebuilds, so the true cost will be somewhat higher.

| Stage                   | Simulated time | Steps             | Estimated wall time |
|-------------------------|----------------|-------------------|---------------------|
| Heating (300 to 2500 K) | 30 ps          | $3.0 \times 10^4$ | $\sim 16$ min       |
| Hold at 2500 K          | 1 ns           | $1.0 \times 10^6$ | $\sim 9$ h          |
| Cooling (1 K/ps)        | 2.2 ns         | $2.2 \times 10^6$ | $\sim 1.7$ h        |

Table 4: Projected cost for 216 atoms on 8 cores (melt-stable for heating and hold, `pruned_1` for cooling).

#### 4.8 The melt

The heating stage ran the 216-atom lattice under `fix npt`, ramping 300 K to 2500 K at 0 bar over 30,000 steps (30 ps), with a thermostat damping of 0.1 ps. The run melted successfully, and two signatures confirmed the liquid. First, the pressure fluctuations grew by an order of magnitude (order  $10^4$  bar vs.  $10^3$  in the solid). Second, the number of neighbour-list rebuilds rose from zero to 400; rebuilds only happen when atoms move far enough to change their neighbours, so this shows the atoms are diffusing rather than vibrating in place. VMD confirms it visually, by showcasing the diamond lattice breaking apart into a disordered liquid (Fig. 4).

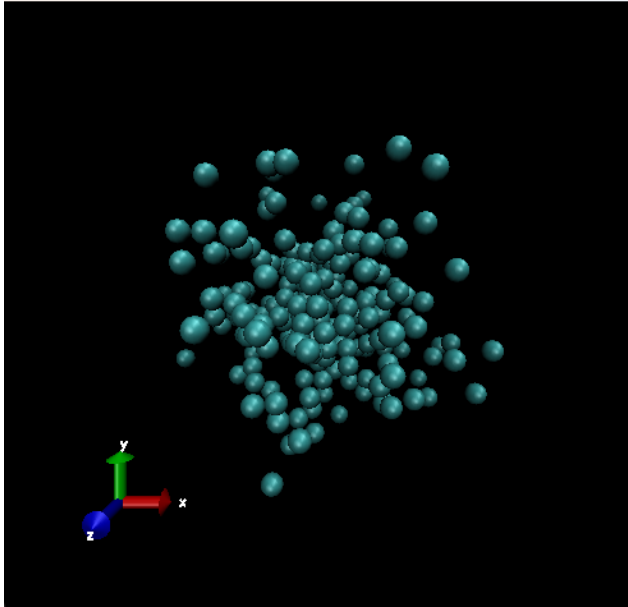


Figure 4: The 216-atom system during the melt, visualized in VMD.

## 4.9 Runs in progress

Three runs are currently executing or queued: (1) the 1 ns hold at 2500 K with `22.almtp`, continuing directly from the saved melt via `read.data`; (2) the same hold with a pruned potential, to test directly whether the pruned potentials survive high temperature, since if they do the hold drops from nine hours to under one; and (3) a second melt with `26.almtp`, to check that the two melt-stable potentials agree.

# 5 Discussion

## 5.1 The toolchain as a deliverable

A large fraction of the elapsed time went into producing two working binaries rather than physics. The supplied build scripts required several environment-specific corrections before compiling on Narval (Table 1), which is a normal part of deploying research software on a shared cluster; the debugging was done with helpful follow-up guidance from Alliance support. The corrected scripts are now documented and reusable by the group, so this cost will not be paid again. The GPU build is validated but unused for silicon so far, which is expected: Meng et al. (2026) intend the GPU variants for roughly 2000+ atoms per GPU, far above 216. It becomes relevant if the system size grows (Section 5.5).

## 5.2 Verifying the atom type mapping

Identifying which atom type corresponds to silicon was a small step, but every later run depends on it. Had the types been assigned the wrong way around, the mistake would not always be obvious: in this case it caused an immediate, unmistakable collapse of the box, but in a system containing both silicon and oxygen, a wrong assignment could instead produce results that look reasonable while being physically wrong. Checking the assignment against a quantity with a known value, here the density of c-Si, is a cheap way to rule this out, and is worth doing for any potential file whose species are not documented.

## 5.3 The potential handoff problem

The  $-12,400$  bar result matters because it is a property of the protocol, not of any one run: the workflow deliberately switches potentials midway, and the equilibrium volume of one is not the equilibrium volume of the other. Two interpretations are possible. One is that this is a technical artifact to be managed: running everything under NPT lets the barostat relax the box to whatever the current potential prefers within a few picoseconds of the switch, which is the approach adopted and also matches the benchmark study. Another possible interpretation is that a disagreement of this size means the pruned potentials describe a measurably different material. Under NPT the disagreement will not appear as an error; it will show up as a difference in the final density of the quenched structure. Since a-Si density ( $2.28 \text{ g/cm}^3$ ) is one of the quantities compared against experiment, this will be tracked explicitly. The paired hold runs are a first test of how far apart the two families actually are.

## 5.4 Feasibility

The cost measurements turned an infeasible protocol into a feasible one: seven days on one core became under two hours for the cooling stage. The remaining bottleneck is the 1 ns hold, the

one stage that currently needs the expensive melt-stable potential. If the pruned potentials prove stable at 2500 K, the whole protocol fits in a single overnight allocation, making it practical to generate an ensemble of independent structures rather than one, which matters because 216-atom statistics from a single sample are unreliable.

## 5.5 System size

The 216-atom size was chosen to match the smallest Zongo et al. models and to keep iteration fast. Its limits are clear: fluctuations scale as  $N^{-1/2}$ , so single measurements are noisy, and a 16 Å box is a small number of correlation lengths for ring statistics and medium-range order. Zongo et al. found crystallinity depends little on system size, which is reassuring. Repeating the protocol at 512 and 4096 atoms is planned for the final weeks to strengthen the statistics, and at those sizes the parallel scaling headroom reopens.

## 5.6 What this enables

The work reported here is almost entirely infrastructure, and it is worth stating plainly what it buys. With the toolchain built and the protocol costed, the remaining effort shifts from software to physics. The quenched structures will support quantitative comparison against experimental a-Si and against the ARTn-MTP models of Zongo et al. (2025), and in particular they will test that paper’s central claim about the melt-quench route: that it tends to trap crystalline grains where ARTn does not. Because the cost is now low enough to generate an ensemble rather than a single structure, that question can be asked statistically rather than anecdotally, and the same structures then feed directly into MAP.

## 5.7 Limitations

- No structural characterization yet: no coordination statistics, bond-angle distributions, ring statistics, or structure factors.
- Cost projections are extrapolated from a static 300 K crystal and will underestimate the liquid stages.
- Melt confirmation is qualitative; a radial distribution function from the melt trajectory should be computed.
- Only `22.almtp` is characterized in any detail, with one cost measurement for `pruned_1`. The other three potentials are untested.

# 6 Status and Future Work

The toolchain is built, validated, and fast enough for the planned protocol. The species mapping, equilibrium volume, and potential costs have been measured, the melt is confirmed, and three runs (two holds and a second melt) are currently executing. The remaining work, in order:

1. **Complete the hold runs** (in progress) and compare the melt-stable and pruned structures, which will show whether the pruned potentials survive 2500 K and how far apart the two families are.
2. **Compute a radial distribution function** from the melt trajectory, to confirm the liquid quantitatively rather than visually.

3. **Run the quench** at 1 K/ps from 2500 K to 300 K under `pruned_1` (about 2 hours on 8 cores), producing the first a-Si structure. The final density will be tracked closely, since the volume disagreement between the potential families is expected to surface there.
4. **Structural analysis:** coordination statistics, bond-angle distribution, ring statistics, and structure factor, compared against Zongo et al. (2025) and experiment, including a check for residual crystallinity.
5. **Apply MAP** to the resulting structures.
6. **Scale up and repeat:** rerun the protocol at 512 and 4096 atoms to strengthen the structural statistics, and generate an ensemble of independent 216-atom structures once the cheapest stable combination of potentials is established.

The main anticipated challenges are the possible instability of the pruned potentials in the melt, which the paired hold runs will settle, and the small system size, which limits the reliability of ring statistics from any single sample.

## A Project Timeline

| Date (2026)  | Milestone                                                              |
|--------------|------------------------------------------------------------------------|
| June 15      | Project start                                                          |
| June 22      | Background reading completed and summarized in research log            |
| June 30      | Lennard-Jones benchmark run and visualized in VMD                      |
| July 1 to 13 | Debugging of the supplied LAMMPS-MTP build scripts                     |
| July 13      | CPU build debugged and validated                                       |
| July 14      | GPU (Kokkos) build debugged and validated                              |
| July 21      | First silicon runs: species mapping, NPT equilibration, NVT production |
| July 22      | Parallel scaling and potential cost measured; volume mismatch found    |
| July 23      | Ensemble and cooling rate settled; first melt launched                 |
| July 24      | Melt confirmed; 1 ns hold and second melt launched                     |
| July 28      | Interim report                                                         |

Table 5: Chronology of the work reported here.

## B Job Submission Scripts

CPU build, Lennard-Jones test:

---

```
#!/bin/bash
#SBATCH --account=ctb-simine
#SBATCH --time=0:15:00
#SBATCH --ntasks=1
#SBATCH --mem=4G

module load gcc openmpi flexiblas fftw eigen voro++
export PATH=$HOME/software/lammps-mtp-cpu/bin:$PATH

lmp -in test.in
```

---

GPU build, same test. Four differences: the `--gres=gpu:1` line, the `cuda` module, the `PATH` pointing to the GPU build, and the Kokkos flags on the `lmp` command.

---

```
#!/bin/bash
#SBATCH --account=ctb-simine
#SBATCH --time=0:15:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --gres=gpu:1
#SBATCH --mem=8G

module load gcc openmpi flexiblas fftw eigen voro++ cuda
```

```
export PATH=$HOME/software/lammps-mtp-gpu/bin:$PATH
```

```
lmp -k on g 1 -sf kk -pk kokkos newton on neigh half -in test.in
```

---

For multi-core runs, LAMMPS is launched through `srun` and memory is requested with `--mem-per-cpu` rather than `--mem`.

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